

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1044

Williams 138 kV
February 25, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1044. The Interconnection Customer proposes a 171.4 MW solar photovoltaic facility interconnecting at the Williams 138 kV substation in the Pennsylvania area. The request was received on August 25, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 344 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,275,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Piney Crk Project – Hilltop 500.0 kV, the Roseland Substation – Ontonagon 138.0 kV line loads to 198.9% of its 179 MVA rating. Project DFAX: 7.2%.

The Chase Hill – Reclaim Substation 138.0 kV is overloaded to 188.8% of its 210 MVA emergency rating for the loss of the Burwell – P.T.S 345.0 kV Line. The project contributes 6.7% DFAX (11.6 MW) to this overload.

For the loss of Burwell – P.T.S 345.0 kV line, the Reclaim – Centerville Rd 138.0 kV Line loads to 179.5% of its 233 MVA rating. Project DFAX: 6.7%.

For the loss of Burwell – P.T.S 345.0 kV, the San Carlos – Centerville Rd 138.0 kV loads to 176.5% of its 244 MVA rating. Project DFAX: 6.7%.

Roseland – Mon Valley Works Illinois 138.0 kV: 175.8% loading (198 MVA) following the outage of the Piney Crk Project – Hilltop 500.0 kV; DFAX = 7.2%.

For the loss of Burwell – P.T.S 345.0 kV, the Somerville – San Carlos 138.0 kV loads to 174.6% of its 252 MVA rating. Project DFAX: 6.7%.

V C Summer – Mon Vly Works Illinois 138.0 kV L: 152.2% loading (218 MVA) following the outage of the Hilltop Substation – Piney Crk Project 500.0 kV; DFAX = 7.2%.

The Palm-Hackberry – Eastside 138.0 kV is overloaded to 148.6% of its 183 MVA emergency rating for the loss of the Sugar Hill – Allegheny Energy Units 3 4 & 5 345.0 kV. The project contributes 6.5% DFAX (11.2 MW) to this overload.

Eastside – N Greenwich 138.0 kV: 147.1% loading (197 MVA) following the outage of the Sugar Hill – Allegheny Energy Units 3 4 & 5 345.0 kV; DFAX = 6.5%.

Country Club – North Greenwich 138.0 kV: 139.5% loading (223 MVA) following the outage of the Sugar Hill – Allegheny Energy Units 3 4 & 5 345.0 kV; DFAX = 6.5%.

For the loss of Sugar Hill – Allegheny Energy Units 3 4 & 5 345.0 kV, the Countryclub Substation – Overton 138.0 kV loads to 139.0% of its 230 MVA rating. Project DFAX: 6.5%.

The Overton – Holland Prk 138.0 kV is overloaded to 137.1% of its 249 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 – Sugar Hill 345.0 kV. The project contributes 6.5% DFAX (11.2 MW) to this overload.

Arsenal – East Kent 138.0 kV: 136.9% loading (224 MVA) following the outage of the Dosamigos – Ontonagon 138.0 kV; DFAX = 11.9%.

N – Holland Park 138.0 kV: 136.9% loading (253 MVA) following the outage of the Sugar Hill – Allegheny Energy Units 3 4 & 5 345.0 kV line; DFAX = 6.5%.

The Sjve El Nido – Palm-Hackberry 138.0 kV is overloaded to 136.2% of its 196 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 – Sugar Hill 345.0 kV. The project contributes 6.5% DFAX (11.2 MW) to this overload.

The Arsenal – Wyandotte Substation 138.0 kV is overloaded to 135.1% of its 239 MVA emergency rating for the loss of the Ontonagon – Dos Amigos Substation 138.0 kV. The project contributes 11.9% DFAX (20.4 MW) to this overload.

The Wyandotte – Williams 138.0 kV L is overloaded to 134.8% of its 246 MVA emergency rating for the loss of the Dos Amigos – Ontonagon 138.0 kV. The project contributes 11.9% DFAX (20.4 MW) to this overload.

For the loss of Ontonagon – Dos Amigos 138.0 kV, the E Kent – Saline 138.0 kV loads to 134.0% of its 195 MVA rating. Project DFAX: 11.9%.

Highland – Sjve El Nido 138.0 kV: 123.7% loading (209 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 – Sugar Hill 345.0 kV line; DFAX = 6.5%.

V C Summer – Spring Ridge Rd 138.0 kV: 123.2% loading (246 MVA) following the outage of the Piney Creek Project – Hilltop 500.0 kV; DFAX = 7.2%.

The Salne – New Holstein Sw Sta 138.0 kV is overloaded to 121.6% of its 208 MVA emergency rating for the loss of the Dos Amigos – Ontonagon Substation 138.0 kV. The project contributes 11.9% DFAX (20.4 MW) to this overload.

For the loss of Hilltop – Piney Creek Project 500.0 kV line, the Spring Ridge Rd Substation – East Omak 138.0 kV loads to 118.7% of its 250 MVA rating. Project DFAX: 7.2%.

The Sta B – Highland 138.0 kV Line is overloaded to 113.9% of its 219 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 – Sugar Hill 345.0 kV L. The project contributes 6.5% DFAX (11.2 MW) to this overload.

For the loss of Ontonagon – Dos Amigos 138.0 kV, the New Holstein Sw Sta – Rainbow Falls Auscble 138.0 kV loads to 112.8% of its 217 MVA rating. Project DFAX: 11.9%.

The Sta Q Harbor – East Omak 138.0 kV L is overloaded to 107.1% of its 260 MVA emergency rating for the loss of the Hilltop – Piney Crk Project 500.0 kV. The project contributes 7.2% DFAX (12.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Roseland Substation – Ontonagon 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,164,000	51% (\$598,000)
Chase Hill – Reclaim 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,310,000	38% (\$866,000)
Reclaim – Centerville Rd 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,752,000	38% (\$1,031,000)
San Carlos – Centerville Rd 138.0 kV	Rebuild line with higher-capacity conductor	\$657,000	38% (\$246,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Roseland – Mon Valley Works Illinois 138.0 kV	Rebuild line with higher-capacity conductor	\$4,987,000	51% (\$2,560,000)
San Carlos – Somerville 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,096,000	38% (\$411,000)
Mon Vly Works Illinois Substation – V C Summer 138.0 kV	Rebuild line with higher-capacity conductor	\$2,206,000	51% (\$1,132,000)
Palm-Hackberry – Eastside 138.0 kV	Rebuild line with higher-capacity conductor	\$1,436,000	100% (\$1,436,000)
Eastside – N Greenwich 138.0 kV	Rebuild line with higher-capacity conductor	\$951,000	100% (\$951,000)
Country Club – North Greenwich 138.0 kV	Rebuild line with higher-capacity conductor	\$3,255,000	100% (\$3,255,000)
Country Club – Overton Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$598,000	100% (\$598,000)
Overton – Holland Park Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$1,809,000	100% (\$1,809,000)
Arsenal – East Kent 138.0 kV	Rebuild line with higher-capacity conductor	\$2,904,000	69% (\$2,003,000)
N – Holland Park 138.0 kV	Rebuild line with higher-capacity conductor	\$958,000	100% (\$958,000)
Palm-Hackberry – Sjve El Nido 138.0 kV	Rebuild line with higher-capacity conductor	\$2,514,000	100% (\$2,514,000)
Wyandotte – Arsenal 138.0 kV	Rebuild line with higher-capacity conductor	\$1,102,000	69% (\$760,000)
Wyandotte – Williams 138.0 kV L	Rebuild line with higher-capacity conductor	\$784,000	69% (\$541,000)
E Kent – Saline 138.0 kV	Rebuild line with higher-capacity conductor	\$1,836,000	69% (\$1,266,000)
Sjve El Nio – Highland 138.0 kV	Rebuild line with higher-capacity conductor	\$6,045,000	100% (\$6,045,000)
V C Summer – Spring Ridge Rd 138.0 kV	Rebuild line with higher-capacity conductor	\$4,346,000	51% (\$2,231,000)
New Holstein Sw Sta – Saline 138.0 kV	Rebuild line with higher-capacity conductor	\$2,292,000	69% (\$1,581,000)
Spring Ridge Rd – East Omak 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,792,000	51% (\$920,000)
Sta B – Highland 138.0 kV Line	Rebuild line with higher-capacity conductor	\$863,000	100% (\$863,000)
New Holstein Sw Sta – Rainbow Falls Auscble 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,872,000	69% (\$2,671,000)
Sta Q Harbor – East Omak 138.0 kV L	Reconductor line	\$2,005,000	51% (\$1,029,000)
	Total allocated to project		\$38,275,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.